

MANGALORE UNIVERSITY

BOTANY

(CORE, ELECTIVE AND SKILL COURSES)

SYLLABUS FOR

B.Sc. UNDER GRADUATE (UG) PROGRAMME

UNDER SEP SCHEME

With effect from

Academic Year 2024-25

Botany

Programme Outcome:

- **Students will be proficient to comprehend classification, morphology, anatomy, and physiology of various groups of plants.**
- **Students will be able to understand the contribution of botany for human welfare with potential uses of plants along with their conservation and sustainable development.**
- **Students will be enriched by various skills related to Gardening and Floriculture, preparation of biofertilizers, mushroom cultivation and ethnobotanical knowledge.**
- **Students will be able to understand and relate physical features of the environment to the structure of population, community, ecosystem, and sustainable conservation strategies.**

COURSES AND CREDITS: B.Sc. with Botany as an Optional Subject

	Semester-wise Course Topics	Teachi ng hrs/ week	Exam duration (hrs)	Marks			No. of Credits
				I.A	Exam	Total	
SEMESTER – I							
Group I	BSCBOCS 101: Diversity of Microbes, Algae and Fungi (T)	4	3	20	80	100	03
	BSCBOPS 101: Diversity of Microbes, Algae and Fungi (P)	4	3	10	40	50	02
SEMESTER – II							
Group I	BSCBOCS 201: Diversity of non-flowering plants and Plant anatomy (T)	4	3	20	80	100	03
	BSCBOPS 201: Diversity of non-flowering plants adPlant anatomy (P)	4	3	10	40	50	02
SEMESTER – III							
Group I	BSCBOCS 301: Angiosperm Morphology and Reproductive Botany (T)	4	3	20	80	100	03
	BSCBOPS 301: Angiosperm Morphology and Reproductive Botany (P)	4	3	10	40	50	02
Group II	BSCBOCES 301: Elective Course: Medicinal Botany	2	2	10	40	50	02
SEMESTER – IV							
Group I	BSCBOCS 401: Plant Taxonomy and Economic Botany (T)	4	3	20	80	100	03
	BSCBOPS 401: Plant Taxonomy and Economic Botany (P)	4	3	10	40	50	02
Group II	BSCBOCES 401: Elective Course: Nursery and Gardening Techniques	2	2	10	40	50	02
SEMESTER – V							
Group I	BSCBOCS 501: Ecology and Conservation Biology (T1)	3	3	20	80	100	03
	BSCBOCS 502: Genetics, Cell and Molecular Biology (T2)	3	3	20	80	100	03
	BSCBOPS 501: Ecology and Conservation Genetics, Cell and Molecular Biology (P)	4	3	10	40	50	02
SEMESTER – VI							
Group I	BSCBOCS 601: Plant Physiology (T1)	3	3	20	80	100	03
	BSCBOCS 602: Plant Breeding and Biotechnology (T2)	3	3	20	80	100	03
	BSCBOPS 601: Plant Physiology& Plant Breeding and Biotechnology (P)	4	3	10	40	50	02
	Skill Enhancement paper - Project work / Field training / Internships						
	T= Theory, P=Practical				Total Credits		46

SEMESTER – I

Diversity of Microbes, Algae and Fungi –Theory

Course Objectives:

- To explore and identify microbes and lower plants, and to gain adequate knowledge on the comparative account of these organisms.
- To impart knowledge about the occurrence, distribution, structure, and life history of microbes and lower plants such as algae, fungi, and lichens.
- To acquaint students with a wide spectrum of plant diseases, their causes, symptoms, and control measures.

Course Outcomes:

After completion of the course, the students will be able to:

- Understand the diversity of microbes in nature.
- Know the diversity of algae, fungi, lichens, and their uses.
- Identify and classify algae and fungi.
- Develop practical skills in staining techniques and slide preparation.
- Identify plant disease symptoms and apply management techniques.

Unit	Topics	Teaching Hours
I	Introduction to Botany: Branches and scope, Major historical developments in Botany including contributions of Indian Botanists (Sir J. C. Bose, P. Maheshwari, B.G.L. Swamy, E.K. Janaki Ammal and M. S. Swaminathan) Career opportunities in Botany. Five kingdom and Three domain systems of classification of organisms with examples Viruses: General Characteristics, Classification based on hosts and nature of genetic material. Ultra- structure and multiplication of TMV and T4 Phage. A brief account of Viroid's and Prions. Bacteria: Types based on cellular morphology, flagellation and mode of nutrition. Ultra structure of a Bacterial cell. Reproduction: binary fission and endospore formation. Genetic recombination in bacteria - conjugation, transformation and transduction (generalized type), Economic importance of Bacteria.	12

II	Algae -1: Occurrence, thallus organization and general methods of reproduction with examples. Pigmentation in algae, Fritsch's classification of algae up to the level of classes with examples. Cyanophyceae: Ultra structure of a cyanobacterial cell and heterocyst. Structure and reproduction of <i>Nostoc</i>. Chlorophyceae: Thallus structure and reproduction of <i>Oedogonium</i> Bacillariophyceae: Types of diatoms with examples- Pennales and Centrales. Thallus structure and reproduction of a Pennate diatom (<i>Pinnularia</i>)	12
III	Algae-2: Phaeophyceae: Thallus structure and reproduction of <i>Sargassum</i> Rhodophyceae: Thallus structure and reproduction of <i>Polysiphonia</i>. Economic importance of algae: Useful aspects- food, SCP, industrial products, medicine, sewage treatment, bio-fertilizers, pollution indicators and energy source. Harmful aspects- algal blooms, algal toxins, and parasitic algae. Fungi: Salient features, occurrence, mycelial organization- prosenchyma, pseudo parenchyma, rhizomorph and sclerotium. General methods of reproduction with example. Alexopoulos system of classification up to the level of classes with examples. Thallus structure and reproduction of <i>Rhizopus</i> (Zygomycetes), <i>Penicillium</i> (Ascomycetes) and <i>Agaricus</i> (Basidiomycetes). Economic importance of Fungi: food value, industrial products, medicinal products, bio control agents and Fungal toxins.	12
IV	Plant Pathology: Introduction, causes and disease management of Sandle spike disease (phytoplasma), Katte disease of Cardamom (virus), Bacterial blight of paddy (bacteria) and Stem bleeding disease of coconut (fungi). Mycorrhizae: definition, types – ecto, endo (VAM) and ectendomycorrhizae, Ecological and economic significance of mycorrhizae. Lichens: Classification- asco and basidio-lichens. Morphological types with examples- crustose, foliose and fruticose. Internal structure of thallus – homeomerous and heteromerous types. Structure of fruiting bodies - soredium, isidium, apothecium and perithecium, Economic importance of lichens.	12
	Total	48 hrs

REFERENCE BOOKS:

1. Suresh Narayan, & Pullaiah, T. (2010). *Eminent Indian botanists – Past and present*. Regency Publications.
2. Dubey, R. C., & Maheshwari, D. K. (2009). *A text book of microbiology*. S. Chand Publishers.
3. Dey, S. N., & Trivedi, P. S. (1977). *A text book of botany: Vol. I*. Vikas Publishing.
4. Gangulee, H. C., Das, K. S., & Datta, C. T. (2002). *College botany: Vol. II*. NCBA (P) Ltd.
5. Kumar, H. D., & Singh, H. N. (1996). *A text book of algae*. East West Press.
6. Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (2008). *Microbiology* (5th ed.). McGraw Hill.
7. Purohit, S. S. (1989). *Viruses, bacteria and mycoplasmas*. Agrobotanical Publications.
8. Singh, V., Pandey, P. N., & Jain, D. K. (2015). *A text book of botany*. Rasogi Publications.
9. Smith, G. M. (1955). *Cryptogamic botany: Vol. I. Algae and fungi* (2nd ed.). McGraw Hill Book Co. Inc.
10. Smith, K. M. (1990). *Plant viruses* (6th ed.). Universal Book Stall.
11. Sundara Rajan, S. (2009). *College botany: Volume I*. Himalaya Publications.
12. Suresh Narayan, & Pullaiah, T. (2010). *Eminent Indian botanists – Past and present*. Regency Publications.
13. Vashistha, B. R., Sinha, A. K., & Singh, V. P. (2004). *Botany for degree students*. S. Chand Publishing.

SEMESTER – I
Diversity of Microbes, Algae and Fungi –Practical

Practical No.	Title of practical exercise/experiment
1	Microscopy technique: Study of compound and Dissection microscopes – parts, working principle, handling and preparation of temporary mountings
2	Microscopic observation of Bacterial cells by simple staining (Positive-Crystal violet, Negative- Nigrosine or Indian Ink). Differential staining of Bacteria - Gram's staining.
3	Study of thallus structure of <i>Nostoc</i> , <i>Oscillatoria</i> and <i>Scytonema</i>
4	Study of thallus structure of <i>Volvox</i> , <i>Oedogonium</i> and <i>Cladophora</i>
5	Study of thallus and reproductive structures of <i>Chara</i> and structures of <i>pennate Diatoms</i>
6	Study of thallus and reproductive structures of <i>Sargassum</i> and <i>Polysiphonia</i>
7	Study of vegetative and reproductive structures of <i>Penicillium</i> , <i>Rhizopus</i> and <i>Puccinia</i>
8	Study of symptoms, causative organism and control measures of Banana Bunchy Top Disease, Citrus Canker and Root Knot of Brinjal, with the help of specimens /photograph.
9	Study of symptoms, causative organism and control measures of Koleroga of Areca Nut, Blast Disease of Rice and Rust of Coffee.
10	Study of Lichens- morphological types, internal structure and reproductive structures
11	Study of local diversity of algae/fungi and preparation of an inventory with photographs/microphotographs (any five not mentioned in practical syllabus)
12	Listing and Study of common fungal/bacterial/viral diseases of local cropplants/plantation crops (any five not mentioned in practical syllabus)

Mangalore University
B.Sc. I Semester -Practical Examination
Diversity of Microbes, Algae and Fungi
Question paper and Scheme of evaluation

Time: 3hrs. Batch..... Date..... Time:am/pm Max. Marks: 40

1. Prepare a stained slide of specimen **A**. Sketch, label and identify with reasons.
Leave the preparation for inspection. 06
2. Identify **B & C** giving reasons, 3+3 = 06
3. Identify **D & E** with reasons and labeled sketches. 3+3 = 06
Prepare a stained slide of the given bacterial sample **F**.
Show the preparation to the examiner and write the procedure. 03
4. Sketch, label and identify the slides **G, H & I** with reasons. 3+3+3 = 09
5. Record Submission and Field Notes. 7+3 = 10

1	A		Prep. Sk Id Cl Reasons 2 1½ ½ ½ 1½ = 6	<u>Reg. No. of Candidates Assigned</u> <u>Reg. No. of Absentees:</u> <u>Total examined:</u> <u>Name and Signature of Examiners:</u> 1. Internal 2. External
2	B		Id. Reasons 1 2 = 3 each	
	C			
3	D		Id Sk. Reasons. ½ 1 1½ = 3 each	
	E			
4	F		Prep - 1 Procedure – 2	
5	G		SkId Cl Reasons 1 ½ ½ 1½ = 3each	
	H			
	I			
6		Record with Field report	7+3 = 10	

Instruction to Examiners:

A - Any algal specimen

B and C- One algal and one fungal specimen

D and E- Any Pathological specimen/Morphological types of lichens

F - Bacterial culture/curd sample/root nodule suspension

G, H and I- Slides (Lichen/Algae/Fungi -one each)

SEMESTER – II

Diversity of Non-Flowering Plants and Plant Anatomy – Theory

Course objectives:

- To study the structure and function of various tissues and their location in the plantbody
- To provide relevant information about Pteridophytes and Gymnosperms along withtheir evolutionary history, their phylogenetic relationships and fossil wealth of the world and economic importance of selected forms.
- To ascertain the importance of fossilization to relate life forms of earlier era

Course Outcome:

On completion of this course, the students will develop the following skills:

- To identify and classify non-flowering plants.
- Will gain basic knowledge of GTS and plant fossils.
- Observation of variations that exist in the internal structure of various parts of a plant andamong different plant groups in support of the evolutionary concept.
- Skill development for the proper description of internal structure using botanical terms, their identification and further classification.
- Induction of the enthusiasm towards the internal structure of locally available plants.

Unit	Topics	Teaching Hours
I	Plant Histology and Anatomy: Introduction to tissues, general classification of tissues. Meristematic tissues: definition, classification based on origin, function, and position. Theory of shoot organization – tunica corpus theory, Theory of root organization - histogen theory. Permanent Tissues: simple permanent tissues - structure, types and functions of parenchyma, sclerenchyma, and collenchyma. Complex permanent tissues - structure, composition and functions of xylem and phloem. Secretory tissues: Glandular trichomes, nectars, glands, laticifers and resin ducts. Primary anatomy of dicot and monocot root, stem, and leaf. Normal secondary growth in dicot stem.	12
II	Bryophytes: Occurrence, General characteristics, Rothmaler's classification up to the level of classes with examples. Thallus morphology, anatomy, and reproduction of <i>Riccia</i> , <i>Anthoceros</i> and <i>Funaria</i> . Evolution of sporophytes and gametophytes in bryophytes – theories of progressive sterilization and simplification. Importance of bryophytes.	12

III	Pteridophytes: Occurrence, General characteristics, G. M. Smith's classification up to the level of classes with examples. Morphology, anatomy and reproduction in <i>Psilotum</i> , <i>Equisetum</i> and <i>Pteris</i> ; Stelar variations in pteridophytes, Heterospory and seed habit. Economic importance of pteridophytes.	12
IV	Gymnosperms: Occurrence, Salient features, Sporne's system of classification up to the level of classes with examples. Morphology, anatomy, and reproduction in <i>Cycas</i> and <i>Gnetum</i> . Economic importance of Gymnosperms. Paleobotany: Introduction, significance of fossils, Geological time scale and Evolution of plant groups, Types of plant fossils – compressions, impressions, incrustations, petrification with examples. Fossil dating- direct and indirect. Fossil Pteridophytes- <i>Rhynia</i> – Morphology of sporophyte. Fossil Gymnosperms- <i>Cycadeoidea</i> - Morphology.	12
	Total	48 hrs

REFERENCE BOOKS

1. **Pandey, B. P.** (2001). *College Botany: Volume I*. S. Chand Publishing. ISBN: 8121905931
2. **Sundara Rajan, S.** (2011). *College botany (Vol. 3)*. Himalaya Publishing House
3. **Kumaresan, V., & Reginald, A.** (2013). *Pteridophytes, gymnosperms and paleobotany*. Saras Publications. ISBN: 9789381927779
4. **Vashishta, P. C.** (2013). *Botany for degree students: Gymnosperms* (Revised ed.). S. Chand Publishing. ISBN: 9788121926188
5. **Johri, B. M.** (2015). *Embryology of angiosperms*. Springer. ISBN: 9788132208591
6. **Sambamurthy, A. V. S. S.** (2005). *Textbook of bryophytes, pteridophytes, gymnosperms and palaeobotany*. J.K. International Publishers. ISBN: 9788187134966

SEMESTER – II
Diversity of Non-Flowering Plants and Plant Anatomy – Practical

Practical No.	Title of exercise /experiments
1	Study of structure and reproduction of <i>Riccia</i> , <i>Anthoceros</i> and any moss
2	Study the morphology, anatomy, and reproductive structure of <i>Selaginella</i> and <i>Equisetum</i>
3	Study the morphology, anatomy, and reproductive structure of <i>Pteris</i> and <i>Marselia</i>
4	Study the morphology, anatomy, and reproductive structure of <i>Cycas</i>
5	Study the morphology, anatomy, and reproductive structure of <i>Pinus</i>
6	Study the morphology, anatomy, and reproductive structure of <i>Gnetum</i>
7	Study of simple and complex permanent tissues
8	Study of dicot and monocot stem (T. S)
9	Study of dicot and monocot root (T. S)
10	Study of Normal secondary growth in dicot stem (T. S)
11	Study of dicot and monocot leaf (T. S)
12	Study of local diversity of Bryophytes / Pteridophytes/Gymnosperms and preparation of an inventory with photographs to be recorded in Record book.

Mangalore University
B.Sc. II Semester - Practical Examination
Diversity of Non-Flowering Plants and Plant Anatomy
Question paper and Scheme of evaluation

Time: 3 hrs. Batch..... Date..... Time... ..am/pm Max. Marks:40

1. Prepare a temporary stained section of the material **A**. sketch, label and identify with reasons. Leave the preparation for inspection 06
2. Identify **B** and **C** with reasons 3+3 = 06
3. Write critical notes on **D** and **E** with labelled sketches 3+3 = 06
4. Identify, sketch and label the slides **F, G, H & I** with reasons 3+3+3+3 = 12
5. Record with field report 7+3 = 10

1	A		Prep. 2	Sk 1½	Id ½	Reasons 2	= 6	Reg. No. of Candidates Assigned
2	B		Id 1	Rea 2			= 3 each	
	C							
3	D		Id ½	Sk 1	Crt.Nts. 1½		= 3 each	
	E							Reg. No. of Absentees:
4	F		Sk 1	Id ½	Reasons 1½		= 3each	Total examined:
	G							Name and Signature of Examiners:
	H							1. Internal
	I							2. External
5		Record with Field report	10					

Instruction to Examiners:

A from dicot or monocot stem/root

B and **C** one specimen each from Bryophyta and Gymnosperms

D and **E** from Pteridophyta

F, G, H and **I** - one slide each from Histology, Bryophyta, Pteridophyta, Gymnosperms

SEMESTER –III

Angiosperm Morphology and Reproductive Botany - Theory

Course objectives:

- To study the morphological variations & modifications in vegetative and floral plants
- To study the reproductive methods and life cycle of angiosperms
- To understand the steps involved in embryogenesis in plants.

Course Outcome: On completion of this course, the students will develop the following skills:

- Observation and classification of the floral variations in different plants.
- Able to understand the various reproductive methods and the life cycle of flowering plants
- Observation and classification of the embryological variations in angiosperms.
- Enthusiasm to understand the evolution based on various methods of reproduction.

Unit	Topic	Teaching Hrs
I	Morphology of Vegetative Structures Parts of a typical flowering plant. Morphology of vegetative organs: Root: Characteristics, types - tap and adventitious, root modification- storage, mechanical support and vital functions. Stem: Characteristics, functions, modification – underground, sub aerial and aerial. Leaf: Parts of typical leaf, venation and modifications of leaf, Phyllotaxy.	12
II	Morphology of Reproductive Structures Inflorescence: Racemose, cymose and special types. Flower: Parts of a typical flower, forms of thalamus, androphore, gynophore, gynandrophore, insertion of floral whorls on the thalamus (hypogyny, perigyny and epigyny), structure, function and modification of calyx, corolla, aestivation, androecium and variations gynoecium and its variations, placentation.	12
III	Reproductive Botany I Microsporangium: Morphology and Anatomy of the mature anther; Anther wall layers; Tapetum -types, structure and functions; sporogenous tissue. Microsporogenesis - Microspore mother cells, microspore tetrads and their types; structure of mature pollen; Pollinia. Microgametogenesis – structure and development of male gametophyte. Pollen embryo sac (Nemec phenomenon). Megasporangium – Ovule and types of ovules - Anatropous, Orthotropous, Amphitropous and Campylotropous. Detailed Structure of Orthotropous ovule. Megagametogenesis – Female gametophyte /embryosac -tenuinucellate and crassinucellate; monosporic - Polygonum type (in detail), bisporic – Allium type, tetrasporic - Fritillaria type (Just mentioning the types in brief). Structure of mature 7-celled embryo sac.	12

IV	Reproductive Botany II Pollination: Definition, self and cross-pollination and their types, contrivances for self and cross-pollination; types of pollination based on agents and their characteristic features. Fertilization: Process and significance of double fertilization, Post fertilization events. Endosperm – Types- Free nuclear (<i>Cocos nucifera</i>), cellular (<i>Cucumis</i>) and helobial. Significance of endosperm. Ruminant endosperm. Embryogenesis: Dicot (<i>Capsella bursa-pastoris</i>) and Monocot (<i>Najas</i>) embryo development. Fruits: Types - simple, aggregate and multiple fruits. Fruit and Seed dispersal strategies. Seed: Structure of Dicot and Monocot seed.	12
	Total	48 hrs

References:

1. Bhojwani, S. S., Bhatnagar, S. P., & Dantu, P. K. (2015). *The embryology of angiosperms* (6th ed., 376 pp.). Vikas Publishing House. ISBN: 9789325981294
2. Pandey, B. P. (2001). *A textbook of botany: Angiosperms* (Illustrated ed., 821 pp.). S. Chand Publishing. ISBN: 9788121904049
3. Bhojwani, S. S., & Soh, W. Y. (Eds.). (2001). *Current trends in the embryology of angiosperms* (544 pp.). Dordrecht: Kluwer Academic Publishers. ISBN: 0-7923-6888-6
4. Eames, A. J. (1961). *Morphology of the angiosperms* (518 pp.). McGraw-Hill.
5. Raghavan, V. (2012). *Developmental biology of flowering plants* (Illustrated ed., 354 pp.). Springer Science & Business Media. ISBN: 9781461212348
6. Saxena, M. R. (1993). *Palynology: A treatise* (127 pp.). New Delhi: Oxford & IBH Publishing Co. Pvt. Ltd.
7. Nair, P. K. K. (1970). *Pollen morphology of angiosperms: A historical and phylogenetic study* (160 pp.). Scholar Publishing House. ISBN: 9780389042044
8. Johri, B. M. (Ed.). (1984). *Embryology of angiosperms* (Illustrated ed., 830 pp.). Springer-Verlag. ISBN: 9780387127392
9. Shukla, A. K., Vijayaraghavan, M. R., & Chaudhry, B. (1998). *Biology of pollen* (133 pp.). APH Publishing. ISBN: 9788170249245
10. Maheshwari, P. (1950). *An introduction to the embryology of angiosperms* (453 pp.). McGraw-Hill.

SEMESTER - III

ANGIOSPERM MORPHOLOGY AND REPRODUCTIVE BOTANY – PRACTICAL

Practical No.	Title of exercise /experiments
1	Morphology and modification of root
2	Morphology and modification of stem
3	Study of inflorescences- Racemose, Cymose and special types
4	Study of fruits- simple, aggregate and multiple fruits
5	Study of flower and its parts; epigynous, perigynous and hypogynous flowers
6	Pollination types and seed dispersal mechanisms (including appendages - aril and caruncle) using photographs and/ specimens).
7	a. Structure of anther (young and mature) using permanent slides. b. Types of placentation and types of ovules (anatropous, orthotropous, campylotropous) using permanent slides.
8	a. Estimation of pollen viability (<i>in vitro</i> method) by hanging drop method. b. Study of pollen grains from any five flowers.
9	Female gametophyte: Polygonum (monosporic) type of Embryo sac Development (Permanent slides/photographs). Ultrastructure of mature egg apparatus cells through electron micrographs.
10	Dissection and mounting of the embryo from developing seeds (monocot and dicot).
11&12	Mini project: Mini project work in groups of 2- 4 students/individual, from the following list. This is to be recorded in the practical record book. i. Leaf morphology (Monocot and Dicot) types and phyllotaxy. ii. Placentation types. iii. Variation in calyx and corolla and their significances. iv. Variation in androecium and gynoecium and their significances. v. Seed dispersal methods. vi. Types of fruits vii. Modification of root/stem/leaves viii. Any topic relevant to the title

Mangalore University
B.Sc. III Semester - Practical Examination
ANGIOSPERM MORPHOLOGY AND REPRODUCTIVE BOTANY
Question paper and Scheme of evaluation

Time: 3 hrs. Batch..... Date..... Time..... am/pm Max. Marks: 40

1. Estimate the percentage of pollen viability in the given material **A** by hanging drop method.
Write the procedure. Leave the preparation for inspection. 07
2. Dissect and mount the embryo from the given material **B** and draw a labelled sketch.
Identify by giving reasons. 04
3. Identify the specimens **C, D and E**, draw a labelled diagram and comment on the morphology/
modification 08
4. Identify the slides **F and G** with labelled diagram and reasons. 08
5. Identify the structure in the given photograph/slide **H** with reasons and labelled sketch. 03
6. Practical Record and project report 10

1	A		Slide - 02, Procedure- 02, Tabulation, Calculation & result- 02 = 06	Reg. No. of Candidates Assigned: Reg. No. of absentees: Total Examined: <u>Examiners:</u> 1. External 2. Internal
2	B		Mounting, Id, Labl sk, Reason 01 01 01 01	
3	C D E		Id, Labl sk, Reason 01 01 01 3x3 = 9	
4	F G		Id, Labl sk, Reason 01 01 02 4x2 = 8	
5	H		Id, Labl sk, Reason 01 01 01 3x1 = 3	
6		Practical Record with Project Report	7+3=10	

Note to the Examiners:

A – Vinca/Impatiens/any suitable flower.

B – Sweet corn (tender seed)/Crotalaria/ Mustard (tender)/ any seeds.

C – Root/Stem modification; & D – Inflorescence, E- Fruit

F & G – Anther, ovule type/Placentation type

H - Electron micrograph of - Ultrastructure of mature egg apparatus/photograph of-pollen grain/ pollen types and dispersion showing ornamentation and aperture.

Elective Course: Semester – III: Medicinal Botany

Course outcome: After the successful completion of the course, the student will be able to:

- Understand the medicinal properties of plants and different systems of medicines
- Understand the adulterations in herbal drugs and evaluation of an herbal drug
- Develop interest in the cultivation and conservation of medicinal plants.

Unit	Topics	Teaching Hours
I	Introduction to medicinal Botany, Role of plants in traditional medical systems - Ayurveda, sidda, unani and folk medicine. Diversity of medicinal plants of India. A medicinally useful plant part with examples – root, stem, leaf, and fruits/seeds. Chemical constituents of herbal drugs with examples- carbohydrates, alkaloids, glycosides, oils, resins, lipids and fibres. Study of morphology, medicinal uses and commercial products of important Indian traditional medicinal plants (List to be given) Aloe vera, Turmeric, Neem, Tulsi, and Zinger.	12
II	Concept of herbal nutraceuticals and cosmeceuticals. Collection and processing of herbal drugs – harvesting, drying, garbling, packing, storage, quality management and documentation. Importance of cultivation and conservation of medicinal plants. Ethnobotany and its significance Adulteration of herbal drugs. Methods of evaluation of herbal drugs – organoleptic, microscopical, chemical, physical, and biological.	12
Total		24 Hours

SUGGESTED REFERENCE BOOKS:

1. Jain, S. K. (2010). *Manual of ethnobotany* (2nd rev. ed.). Scientific Publishers. ISBN: 9387307859
2. Dutt, A. (2009). *An introduction to medicinal plants*. Adhyayan Publishers and Distributors. ISBN: 978-8184350623
3. Atal, C. K., & Kapoor, B. M. (1982). *Cultivation and utilization of medicinal plants*. Regional Research Laboratory, Jammu Tawi.
4. Shah, B., & Seth, A. K. (2014). *Textbook of pharmacognosy and phytochemistry*. Elsevier. ISBN: 9788131234737
5. Trivedi, P. C. (2009). *Medicinal plants: Utilization and conservation*. Avishkar Publishers. ISBN: 8179102857
6. Akerele, O., Heywood, V., & Synge, H. (1991). *The conservation of medicinal plants*. Cambridge University Press. ISBN: 9780521392005
7. Chaudhary, A. B. (2007). *Endangered medicinal plants*. Daya Publishing House. ISBN: 8170354605
8. Kokate, C. K., Purohit, A. P., & Gokhale, S. B. (2018). *Pharmacognosy* (58th ed.). Nirali Prakashan. ISBN: 9789387669123

SEMESTER – IV

Taxonomy and Economic Botany –Theory

Course objectives:

- To study the methods of identification, classification, and nomenclature of angiosperms.
- To know economically important plants and their uses.

Course outcome:

After the successful completion of the course, the student will be able to:

- Understand the different systems of plant classification
- Identify, classify, and describe a plant in scientific terms, thereby, identification of plants using dichotomous keys, skill development in identification and classification of flowering plants.
- Interpret the rules of ICN in botanical nomenclature
- Recognize the importance of herbarium and digital herbarium,
- Recognize locally available angiosperm families and plants of economic importance.

Unit	Topics	Teaching Hours
I	Introduction to Taxonomy: History, objectives and scope of Taxonomy Systems of classification: Artificial - Carolus Linnaeus's, Natural- Bentham and Hooker's, Phylogenetic - Engler and Prantl's, their merits and demerits. A brief study of APG system of plant classification with focus to latest APG IV system and its significance Plant identification: Taxonomic dichotomous keys; brief account of intended (yoked) and bracketed keys. Botanical nomenclature: Binomial nomenclature, a brief introduction to ICBN- principles and rules, ranks of taxa and taxonomic types. Brief introduction to cyto-taxonomy, chemo-taxonomy, numerical taxonomy and molecular taxonomy, DNA barcoding of plants.	12
II	Herbaria and herbarium techniques, digital herbaria. Botanical Survey of India, Botanical gardens, and their functions. Distinguishing features with economic importance of any five local examples from the following families of angiosperms, according to Bentham and Hooker system: Polypetalae- Malvaceae, Fabaceae and its sub-families, Rutaceae, Anacardeaceae, Cucurbitaceae and Apiaceae.	12
III	Distinguishing features with economic importance of any five local examples from the following families of angiosperms according to Bentham and Hooker system: Gamopetalae- Rubiaceae, Asteraceae, Apocynaceae, Solanaceae and Lamiaceae. Monochlamydeae- Amaranthaceae, Euphorbiaceae and Moraceae Monocotyledonae- Orchidaceae, Arecaceae and Poaceae.	12

IV	Economic Botany: Introduction to economic botany: Definition and significance Cereals: Botany*, products and uses of Rice and Wheat. Brief account of millets. Pulses/ Legumes: Botany and uses of Mung bean and Pigeon pea Brief account of fodder legumes. Spices and condiments: Definition. Botany and uses of Cinnamon, Clove, Black pepper, Cardamom, Ginger, and Coriander. Beverages: Definition. Botany*, processing and uses of Tea and Coffee Oil yielding plants: Types of plant oils with examples -fatty oils and essential oils. Botany*, extraction/processing and uses of Coconut and Sandalwood oil. Fiber yielding plants: Classification based on origin of fibers. Morphology, extraction and uses of Cotton and Coir fibers. Timber yielding plants: Important timber yielding plants of India (any five) and their uses. Rubber yielding plants: Botanical source, extraction/processing, and uses of Hevea rubber Medicinal plants: Botany* and uses of <i>Rauwolfia serpentina</i>, <i>Centella asiatica</i>, <i>Tinospora cordifolia</i> and <i>Aloe vera</i> A general account of outdoor and indoor ornamental plants. A general account of ethnobotany: definition and its significance. Note: *Botany includes botanical name, family, place of origin and parts used	12
	Total	48 hrs

Reference Books:

1. **Jain, S. K.** (2010). *Manual of ethnobotany* (2nd rev. ed.). Scientific Publishers. ISBN: 9387307859
2. **Datta, S. C.** (1989). *Systematic botany*. New Age International. ISBN: 8122400132, 9788122400137
3. **Singh, G.** (2019). *Plant systematics: Theory and practice* (3rd ed.). CBS Publishers & Distributors Pvt. Ltd. ISBN: 9788120417632.
4. **Soundar Rajan, S.** (2007). *College botany Vol. V, Part I: Taxonomy and economic botany*. Himalaya Publishing House.
5. **Mukherjee, S. K.** (2004). *College botany (Vol. III)*. New Central Book Agency.
6. **Pandey, B. P.** (1999). *Economic botany* (5th ed.). S. Chand Publication. ISBN: 978-8121903417
7. **Bhat, K. G.** (2014). *Flora of South Kanara: Dakshina Kannada and Udupi districts of Karnataka*. Taxonomy Research Centre. ISBN: 9789383765126
8. **Bhat, K. G.** (2014). *Flora of Udupi*. Taxonomy Research Centre. ISBN: 9788190251413
9. **Singh, V.** (1981). *Taxonomy of angiosperms*. Rastogi Publications. ISBN: 9788171338498
10. **Kochhar, S. L.** (2011). *Economic botany in the tropics* (4th ed.). Macmillan Publishers India Ltd.

SEMESTER – IV

Taxonomy and Economic Botany -Practical

Practical No.	Experiments
1	Study of Malvaceae
2	Study of Fabaceae sub- families
3	Study of families Myrtaceae and Apiaceae
4	Study of families Rubiaceae and Asteraceae
5	Study of families Apocynaceae and Lamiaceae
6	Study of families Amaranthaceae and Euphorbiaceae
7	Study of families Arecaceae and Poaceae
8 & 9	Botany and uses of economically important plants with observation of plants/plant products listed as follows: Rice, wheat, Ragi, Bengal gram Cinnamon, Clove, Black pepper, Cardamom, Ginger and Coriander, Tea powder, Coffee powder, Cocoa fruit, Cotton, Coir and Jute fibers, Rubber, Jaggery, Brahmi (<i>Centella asiatica</i>) and <i>Aloe vera</i> .
10	Study of outdoor and indoor ornamental plants with photographs (two each).
11 & 12	Mini project report in groups of 2- 4 students, from any one of the following lists to be recorded in the practical record. a) Study of the members of the family Cucurbitaceae (any 5) b) Study of the members of the family Anacardiaceae (any 5) c) Study of Orchid flower. d) Preparation of digital herbarium for plant specimens (any 2) e) Chart preparation on processing/extraction of any one essential oil f) Any relevant topic

Activities:

Field visit to study the botanical name, family, common names and economic/ ethno-botanical importance of local angiosperms. A brief report of such field visit with the list of plants observed and field photographs/diagrams should be included in the practical record.

Mangalore University
B.Sc. IV Semester- Practical Examination
TAXONOMY and ECONOMIC BOTANY
Question Paper and Scheme of evaluation

Time: 3hrs. Batch..... Date..... Time... ..am/pm Max. Marks: 40

- | | |
|--|-----------|
| 1. Derive and describe the specimen A, B & C taxonomically (3 x 4) | 12 |
| 2. Explain the specimen D using technical terms with the family and Botanical name | 05 |
| 3. Write the floral diagram and floral formula of the given specimen E | 05 |
| 4. Write botanical name, common name, family and part used of F, G, H & I (4 x 2) | 08 |
| 5. Practical Record with field visit and mini project report | 10 |

1	A B C		Derivation 01 Family and Botanical name 01 Description 02 04 3x4 = 12	Reg. No. of Candidates Assigned
2	D		Family and Botanical name 01 Technical description 04 05	Reg. No. of absentees:
3	E		Floral formula 02 Floral diagram 03 05	
4	F G H I		Botanical name ½ Common name ½ Family name ½ Uses/Significance ½ 4x2 = 8	1. External
5		Practical Record with field visit &Project report	10	2. Internal

Note:

A B & C Each from Polypetalae, Gamopetalae and Monochlamydeae

D Plant specimen with inflorescence/flower from any of the family studied.

E Plant Specimen from any of the family studied.

F, G, H Plant/plant product from any of the specimen mentioned in the practical syllabus.

I - Indoor/Outdoor ornamental plant

Elective Course: Semester - IV

Nursery and Gardening Management

Course outcome: After the successful completion of the course, the student will be able to:

- To introduce students to the principles and practices involved in nursery and garden management.
- To develop skills in planning, design, establishment, and maintenance of nurseries and gardens
- Understand the nursery techniques and management
- Develop the knowledge of different types of gardens and garden plants and their management

Unit	Content of Theory Elective Course	Teaching Hours
I	Nursery: Definition, Objectives, and Scope of Nursery. General Practices: Seed collection, sowing, watering, weeding, fertilization, pest and disease control. Infrastructure Requirements: Site selection, layout planning, fencing, irrigation systems, storage, work sheds. Seasonal Planning and Activities: Calendar-based nursery activities—sowing, transplanting, pruning, etc. Planting Techniques: Direct seeding vs. transplanting methods. Growth Media: Soil-based and soilless media (e.g., cocopeat, perlite, vermiculite). Synthetic growth media for pots and nursery beds. Hardening of Plants: Methods to acclimatize nursery plants before field planting. Protected Structures: Greenhouse, Mist chamber, Shade net house / Shed roof, Glasshouse. Bonsai Techniques: Cultural requirements, Tools and materials, Art and aesthetics of bonsai creation and maintenance.	12hrs
II	Gardening: Definition, Objectives, and Scope of Gardening. Principles of Gardening: Unity, balance, simplicity, rhythm, and proportion. Garden Components: Paths, hedges, fences, gates, seating, arches, pergolas. Adornments and Accessories: Garden lights, sculptures, birdbaths, fountains. Lawn Making and Maintenance: Site preparation, Grass selection, Sowing/turfing, Irrigation and mowing. Design of Special Features: Rockery, Water garden. Types of Gardens: Vertical gardens, Roof gardens, Public parks and gardens, Indoor gardening, Therapeutic/Healing gardens, Japanese gardens, Mughal gardens, Zen gardens (as special types).	12hrs
	Total	24 Hours

Text Books and References

1. Agrawal, P.K. (1993). Hand Book of Seed Technology. New Delhi, Delhi: Dept. of Agriculture and Cooperation, National Seed Corporation Ltd.
2. Bose T.K., Mukherjee, D. (1972). Gardening in India. New Delhi, Delhi: Oxford & IBH Publishing.
3. Jules, J. (1979). Horticultural Science, 3rd Edition. San Francisco, California: W.H. Freeman and Co.
4. Kumar, N. (1997). Introduction to Horticulture. Nagercoil, Tamil Nadu: Rajalakshmi Publications.
5. Musser E. Andres. (2005). Fundamentals of Horticulture. New Delhi, Delhi: Mc Graw Hill Book Co.

SEMESTER V – Paper V

BSCBOCS 501 - Ecology and Conservation Biology – Theory

Course Objectives

- To understand the interactions between organisms and their environment.
- To create awareness of ecological principles and contemporary environmental issues.
- To develop an understanding of biodiversity conservation and sustainable management of natural resources.

Course Outcomes

After successful completion of the course, the student will be able to:

- Explain the influence of biotic and abiotic factors on organisms and ecosystems.
- Describe the structure, functioning, and stability of ecosystems.
- Apply the principles of biodiversity conservation and environmental management for sustainable development.

Unit	Topics	40 Hours
I	Fundamentals of Ecology and Ecological Adaptations: Levels of ecological organization. Ecological factors: climatic factors – light, temperature, precipitation and its effect on plants; edaphic factors – soil as an ecological factor: soil texture, soil pH, soil aeration, organic matter, humus and soil microorganisms. Ecological groups of plants and their adaptations: hydrophytes, xerophytes, epiphytes and halophytes (morphological and anatomical adaptations).	10
II	Ecosystem and Plan succession: Concept, structure and functions of ecosystems. Types of ecosystems – terrestrial (forest) and aquatic (pond) ecosystems. Ecosystem processes: Food chain, food web, ecological pyramids and energy flow. Ecological succession: definition, types (primary and secondary succession) and general stages of succession- Hydrosere and Xerosere. Concept of ecological niche, ecotone and ecotypes.	10
III	Phytogeography, Biodiversity and Conservation: Phytogeographical regions of India. Vegetation types of Karnataka – evergreen, semi-evergreen, deciduous, scrub, mangroves and shola forests. Biodiversity: definition and types (habitat, species and genetic diversity), threats to biodiversity -habitat loss, invasive alien species and climate change. Concept of endemism and endemic species. Biodiversity hotspots of India with special reference to the Western Ghats. IUCN categories and Red List plant species of Karnataka. Conservation methods: in-situ conservation (biosphere reserves, national parks and sacred groves) and ex-situ conservation (botanical gardens, seed banks, gene banks, pollen banks, culture collections and cryopreservation).	10
IV	Environmental Pollution, Climate Change and Sustainability: Water pollution – sources, eutrophication, emerging contaminants (microplastics and pharmaceutical residues), effects and control measures. Air pollution	10

	– major air pollutants, particulate matter (PM_{2.5} and PM₁₀), greenhouse gases, effects and mitigation measures. Soil pollution – heavy metals, pesticide residues, solid waste and e-waste, effects and remediation measures including phytoremediation. Climate change and global warming: causes, effects on agriculture and mitigation strategies. Environmental sustainability: solid waste management, Sewage treatment, Biofuels.	
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REFERENCES:

- Bendre, A. M., & Pande, P. C. (2006). *Introductory botany*. Rastogi Publications.
- Chapman, J. L., & Reiss, M. J. (1992). *Ecology: Principles and applications*. Cambridge University Press.
- Kaushik, A., & Kaushik, C. P. (1990). *Perspectives in environmental studies*. New Age International Publishers.
- Kochhar, P. L. (1975). *Plant ecology* (9th ed.). New Delhi Bombay.
- Kumar, H. D. (1992). *Modern concepts of ecology* (7th ed.). Vikas Publishing House.
- Mohan, P. A. (2002). *Ecology*. Himalaya Publishing House.
- Odum, E. P. (1975). *Ecology*. Holt, Rinehart and Winston.
- Sharma, P. D. (2018). *Fundamentals of ecology*. Rastogi Publications.
- Shukla, R. S., & Chandel, P. S. (2005). *A textbook of plant ecology, ethnobotany and soil science* (10th ed.). S. Chand Publishing.
- Verma, P. S., & Agarwal, V. K. (2010). *Cell biology, genetics, molecular biology, evolution and ecology*. S. Chand & Company Pvt. Ltd.
- Weaver, J. E., & Clements, F. E. (1938). *Plant ecology*. Tata McGraw-Hill Publishing Company Ltd.

SEMESTER – V; Paper – VI
BSCBOCS 502 - Genetics, Cell and Molecular Biology – Theory

Course Objectives

- To understand the structure and functions of the cell and its organelles.
- To study cell division and its significance in plant growth and development.
- To understand the principles of inheritance in plants.
- To study the molecular basis of plant life, including nucleic acids and gene expression.

Course Outcomes

After successful completion of the course, the student will be able to:

- Explain the structure and functions of the cell and its organelles.
- Differentiate between mitosis and meiosis and describe their roles in plant growth, development, and reproduction.
- Explain the basic principles of inheritance and patterns of genetic transmission in plants.
- Describe the molecular organization of plants, including the structure and functions of DNA, RNA, and proteins.

Unit	Topics	40 Hours
I	Cell Biology and Cytology: Structure of plant cell. Ultra-structure and functions of cell wall, plasma membrane (fluid mosaic model), endoplasmic reticulum, chloroplast, ribosomes, Golgi complex, mitochondria, peroxisomes, plastids and vacuole. Ultra-structure of nucleus – nuclear envelope, nuclear pore complex and nucleolus. Ergastic substances in plant cells. Introduction to cell signalling. Transport across membranes – diffusion, osmosis and active transport.	10
II	Cell Division and Chromosome Biology: Types of chromosomes based on centromere position; autosomes and allosomes. Structure of metaphase chromosome; nucleosome model of chromosome organization. Cell cycle and its phases. Mitosis in plant cells – karyokinesis and cytokinesis; significance of mitosis. Meiosis in plant cells; significance of meiosis. Variations in Chromosome Number: Euploidy and aneuploidy; autopolyploidy and allopolyploidy; monosomy, trisomy and nullisomy with suitable plant examples. 1 Chromosomal Aberrations: Structural chromosomal aberrations – deletion, duplication, inversion and translocation	10
III	Molecular Biology and Gene Expression: Experiments proving DNA as the genetic material – Griffith, Avery–MacLeod–McCarty and Hershey–Chase experiments. DNA: chemical composition, Watson and Crick model, semiconservative replication. RNA: structure and functions of mRNA, tRNA and rRNA. Structure of gene – cistron, recon and muton concepts; prokaryotic and eukaryotic genes.	10

	Genetic code and its properties. Gene expression in prokaryotes – Lac operon model. Gene expression in eukaryotes – transcription, RNA processing (gene splicing and post-transcriptional modifications) and translation.	
IV	Principles of Genetics: Introduction to genetics and Mendel's experiments. Monohybrid and dihybrid crosses; laws of segregation and independent assortment. Test cross and back cross – monohybrid and dihybrid. Incomplete dominance – flower colour in <i>Mirabilis jalapa</i> . Multiple alleles and self-incompatibility in tobacco. Gene interactions: complementary (9:7), supplementary (9:3:4) and dominant epistasis (12:3:1). Polygenic inheritance – kernel colour in wheat. Linkage and its types – cis and trans, complete and incomplete linkage; linkage in maize. Gene mutations – spontaneous and induced mutations; physical and chemical mutagens.	10

REFERENCES:

- Burns, G. W. (1983). *The science of genetics: An introduction to heredity* (5th ed.). Macmillan Publishers.
- Cooper, G. M., & Hausman, R. E. (2009). *The cell: A molecular approach* (5th ed.). ASM Press & Sinauer Associates.
- De Robertis, E. D. P., & De Robertis, E. M. F. (2006). *Cell and molecular biology* (8th ed.). Lippincott Williams & Wilkins.
- Freifelder, D. (1996). *Essentials of molecular biology*. Panima Publishing Company.
- Gardner, E. J., Simmons, M. J., & Snustad, D. P. (1991). *Principles of genetics* (8th ed.). John Wiley & Sons.
- Gardner, E. J., Simmons, M. J., & Snustad, D. P. (2008). *Principles of genetics* (8th ed.). Wiley India.
- Griffiths, A. J. F., Wessler, S. R., Lewontin, R. C., & Carroll, S. B. (2012). *Introduction to genetic analysis* (9th ed.). W. H. Freeman & Company.
- Gupta, P. K. (1999). *A textbook of cell and molecular biology*. Rastogi Publications.
- Gupta, P. K. (2000). *Genetics and cytogenetics*. Rastogi Publications.
- Klug, W. S., & Cummings, M. R. (2003). *Concepts of genetics* (7th ed.). Pearson Education.
- Kumar, H. D. (2000). *Molecular biology*. Vikas Publishing House.
- Malacinski, G. M., & Freifelder, D. (1998). *Essentials of molecular biology* (3rd ed.). Jones & Bartlett Publishers.
- Powar, C. B. (2005). *Cell biology* (3rd ed.). Himalaya Publishing House.
- Sambamurty, A. V. S. S. (1999). *Genetics*. Narosa Publishing House.
- Sundara Rajan, S. (2004). *Genetics*. Anmol Publications.

SEMESTER –V

Ecology, Genetics, Cell and Molecular Biology– Practical

Practical No.	Experiments
1	Hydrophytes: Morphological adaptations in <i>Pistia</i> , <i>Eichhornia</i> , <i>Hydrilla</i> , <i>Nymphaea</i> and Anatomical adaptations of Hydrophytes - <i>Hydrilla</i> and <i>Nymphaea</i> .
2	Xerophytes: Morphological adaptations in <i>Asparagus</i> , <i>Casuarina</i> , <i>Acacia</i> , <i>Aloe vera</i> , <i>Euphorbia tirucalli</i> and Anatomical adaptations of <i>Casuarina phylloclade</i> / <i>Acacia phylloclade</i> .
3	Epiphytes: Morphological adaptations in <i>Acampe</i> , <i>Bulbophyllum</i> , <i>Drynaria</i> and Anatomical adaptations of epiphytic root of <i>Acampe</i> / <i>Vanda</i> . Halophytes: Vivipary and Pneumatophores
4	Demonstration of different types of vegetation sampling methods – transects and quadrats. Field visits to study different types of local vegetations/ecosystems and the report to be written in practical record book
5	Project work Ecology
6	Project work
7	Study of ergastic substances - starch (potato), aleurone (wheat grain), calcium oxalate (<i>Pistia</i>), calcium carbonate (<i>Ficus</i> leaf), oil globules (castor seed) and Raphides (<i>Colocasia</i> petiole). Observation of permanent slides of mitosis and meiosis
8	Study of different stages of mitosis by squash technique using Onion root tip
9	Study of different stages of meiosis using permanent slides.
10	Measurement of length and breadth of cells by micrometry technique (Using Onion or <i>Rhoeo</i> leaf cells)
11	Genetic problems related to Mendel's law and incomplete dominance (any Four)
12	Genetic problems related to interaction of genes (any Four)

Mangalore University
III B. Sc. Semester - V, Practical Examination
Ecology, Genetics, Cell and Molecular Biology
Question paper and Scheme of evaluation

Time: 3 hrs. Batch..... Date..... Time.....am/pm Max. Marks: 40

1. Prepare a stained temporary mount of the T.S. of material **A**. Draw a neat labeled diagram. 05
 Leave the slide for inspection. 05
2. Prepare a squash of **B** for the study of mitosis. Leave the slide for inspection. 06
3. Identify the ecological group of **C** and **D**. Comment on their ecological adaptation. 04
4. Identify the slides **E & F** and Comment on it. 06
5. Solve the genetic problem **G** 04
6. Practical Record with field visit report 10
7. Project work report 05

1	A		Prep. 02 Sk 01 Features <u>02</u> 05	Reg. No. of Candidates Assigned: <u>Reg. No. of</u> <u>absentees:</u> <u>Total examined:</u> <u>Examiners:</u> 1.External 2.Internal
2	B		Prep. & Stage 03 Sk 01 Comments <u>02</u> 06	
3	C		Ecol.Group 01 Ecol.Features <u>01</u>	
	D		02 each	
4	E		Identification 01 Sk 01	
	F		Comment <u>01</u> 03 each	
5	G		Problem solve 03 Conclusion <u>01</u>	
6		Practical Record with field visit report	10	
7		Project report	05	

Instruction to Examiners:

A from Hydrophyte/Xerophyte/ Epiphyte

B Mitosis squash

C and **D** Specimen from Hydrophyte/Xerophyte/ Epiphyte/Halophyte

E- Stages of Meiosis and **F**-Ergastic substance

G -Any one genetic problem mentioned in the practical syllabus

SEMESTER – VI; Paper – VII

BSCBOCS 601 - Plant Physiology – Theory

Course Objectives

- To understand the major physiological processes occurring in plants.
- To study the mechanisms and significance of various physiological processes in plant growth and development.
- To understand the role of plant metabolites in metabolism, growth, and adaptation.

Course Outcomes

After successful completion of the course, the student will be able to:

- Explain the major physiological and metabolic processes in plants.
- Analyze the factors influencing plant physiological activities and their responses to environmental conditions.
- Describe the roles of primary and secondary metabolites in plant growth, development, and adaptation.

Unit	Topics	40 Hours
I	Plant Water Relations and Mineral Nutrition: Plant water relations: mechanism of water absorption and factors affecting water absorption. Ascent of sap: cohesion-tension theory. Transpiration: types, significance and factors affecting transpiration; mechanism of stomatal movement (K^+ ion exchange mechanism); guttation. Translocation of organic solutes: source-sink relationship and pressure flow (mass flow) hypothesis. Mineral nutrition: essential elements – macro and micronutrients; functions and deficiency symptoms of N, P, K, Mg, Zn, Mn and B. Hydroponics and its applications.	10
II	Photosynthesis, Enzymes: Photosynthesis: photosynthetic pigments, Photosystem I and II, electron transport and photophosphorylation. C_3 , C_4 and CAM pathways; photorespiration; comparison of C_3 and C_4 plants. Factors affecting photosynthesis. Bioenergetics: concept and significance. Enzymes: classification, properties, mechanism of enzyme action – lock and key, induced fit model, factors affecting enzyme activity.	10
III	Respiration and Metabolism of Biomolecules: Respiration: glycolysis, TCA cycle, oxidative phosphorylation and ATP synthesis; anaerobic respiration (alcoholic and lactic acid fermentation); respiratory quotient. Carbohydrate metabolism: synthesis and breakdown of starch and sucrose. Nitrogen metabolism: biological nitrogen fixation and nitrate reduction. Lipid metabolism: β -oxidation and glyoxylate cycle. Introduction to plant stress physiology: drought, salinity and temperature stress. Secondary metabolites: Alkaloids, terpenoids, phenolics, flavonoids and glycosides with examples.	10
IV	Plant Growth and Development: Plant growth regulators: classification and physiological roles of auxins, gibberellins, cytokinins, abscisic acid and ethylene. Plant movements: tropic and nastic movements. Seed dormancy: causes and methods of breaking dormancy. Photoperiodism and flowering:	10

	short-day, long-day and day-neutral plants; phytochrome, photoperiodic induction and vernalization.	
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References

- Jain, V. K. (2008). *Fundamentals of plant physiology*. S. Chand & Company.
- Kochhar, P. L., & Krishnamoorthy, H. N. (n.d.). *Plant physiology*. Atma Ram & Sons.
- Kumar, A., & Purohit, S. S. (n.d.). *Plant physiology: Fundamentals and applications*. Agrobotanical Publishers.
- Malik, C. P. (2002). *Plant physiology*. Kalyani Publishers.
- Mukherji, S., & Ghosh, A. K. (2005). *Plant physiology*. New Central Book Agency.
- Noggle, G. R., & Fritz, G. J. (n.d.). *Introductory plant physiology*. Prentice Hall of India.
- Pandey, S. N., & Sinha, B. K. (2006). *Plant physiology*. Vikas Publishing House.
- Salisbury, F. B., & Ross, C. W. (1992). *Plant physiology*. CBS Publishers & Distributors.
- Wilson, K., & Walker, J. (1994). *Fundamentals of biochemistry* (2nd ed.). John Wiley & Sons.

SEMESTER – VI; Paper – VIII
BSCBOCS 602 - Plant Breeding and Biotechnology

Course Objectives

- To understand the principles and methods of plant breeding for crop improvement.
- To study the techniques and applications of plant tissue culture in plant propagation and conservation.
- To understand the basic concepts and applications of plant biotechnology in agriculture.

Course Outcomes

After successful completion of the course, the student will be able to:

- Explain the principles and methods of plant breeding and their role in crop improvement.
- Demonstrate an understanding of plant tissue culture techniques and their applications in plant propagation and conservation.
- Describe the tools, techniques, and applications of plant biotechnology, including genetic engineering and transgenic plants.

Unit	Topics	40 Hours
I	Principles of Plant Breeding: History, objectives and achievements of plant breeding. Centres of origin of crop plants. Plant genetic resources, germplasm conservation and gene banks. Genetic basis of breeding self-pollinated and cross-pollinated crops. Methods of plant breeding: mass selection, pure line selection and clonal selection. Hybrid breeding – heterosis, inbreeding and hybridization techniques. Introduction to molecular markers in plant breeding.	10
II	Crop Improvement and Modern Plant Breeding: Polyploidy breeding and mutation breeding. Self-incompatibility and male sterility in crop plants. Development and release of cultivars. Seed production and certification. Plant Breeders' Rights and Farmers' Rights. Role of plant breeding in food security, biofortification and climate-resilient crop development.	10
III	Plant Tissue Culture: History, scope and concept of cellular totipotency. Plant tissue culture laboratory and equipment. Sterilization, culture media preparation and role of plant growth regulators. Types of cultures – callus culture, organ culture and organogenesis. Pollen culture and haploid plant production. Embryo culture, somatic embryogenesis and synthetic seeds. Micropropagation and its commercial applications. Applications of plant tissue culture in agriculture, horticulture and plant conservation.	10
IV	Plant Biotechnology and Genetic Engineering: Introduction to plant biotechnology and genetic engineering. Tools used in genetic engineering – restriction endonucleases, DNA ligase and vectors (plasmids and Ti plasmid). Principles and steps of recombinant DNA technology. Gene transfer methods – Agrobacterium-mediated transformation and particle bombardment. Plant regeneration from transformed cells. Transgenic plants – concept and examples. Golden Rice and Bt cotton. Genome editing using CRISPR-Cas systems (brief account). Biosafety and bioethics of genetically modified crops.	10

REFERENCES

- Acquaah, G. (2012). *Principles of plant genetics and breeding*. Wiley-Blackwell.
- Allard, R. W. (1999). *Principles of plant breeding*. John Wiley & Sons.
- Bhojwani, S. S., & Razdan, M. K. (1996). *Plant tissue culture: Theory and practice*. Elsevier Science.
- Bose, T. K., & Mukherjee, D. (1972). *Gardening in India*. Oxford & IBH Publishing.
- Glick, B. R., & Pasternak, J. J. (2003). *Molecular biotechnology: Principles and applications of recombinant DNA*. ASM Press.
- Gupta, P. K. (2010). *Plant biotechnology*. Rastogi Publications.
- Jain, M. (2014). *Environmental biotechnology*. Narosa Publishing House.
- Kumar, N. (1997). *Introduction to horticulture*. Rajalakshmi Publications.
- Mohapatra, P. K. (2007). *Textbook of environmental biotechnology*. IK International Publishing House.
- Pullaiah, T., & Subba Rao, M. V. (2009). *Plant tissue culture*. Scientific Publishers.
- Sandhu, M. K. (1989). *Plant propagation*. Wiley Eastern Ltd.
- Stewart, C. N., Jr. (2008). *Plant biotechnology and genetics: Principles, techniques, and applications*. John Wiley & Sons.

SEMESTER –VI

Plant Physiology and Plant Breeding & Biotechnology

Practical No	Experiments
	Major experiments
1	Experiment to measure the solute potential of a cell sap by plasmolytic method.
2	Demonstration of transpiration pull/Suction force due to transpiration.
3	Separation of plant pigments by paper chromatography method.
4	Demonstration of CO ₂ liberation during respiration using Ganong's respiroscope
	Minor Experiments
5	Potato osmoscope /Thistle funnel experiment to demonstrate osmosis Demonstration of imbibition pressure using germinating seeds
6	Demonstration of O ₂ liberation during photosynthesis by bubbling method Demonstration of fermentation using Kuhne's Fermentation vessel
7	Demonstration of plant movements – Heliotropism, Geotropism, Hydrotropism Measurement of growth using arc auxanometer
8	Qualitative test for carbohydrates, Protein, fat, and oil
9	Study of plant propagation methods-Cutting, Layering and Grafting
10	Techniques of emasculation, artificial pollination, and bagging
11(a)	Study of Biotechnology lab equipment's – Autoclave, Laminar air flow cabinet, Hot air oven, Incubator, Colony counter, Hemocytometer
11 (b)	Preparation of Synthetic seeds by alginate encapsulation
12	Visits to local nurseries and institutions involved in plant breeding to study the various methods and techniques employed/ Biotechnology Lab or institutions/biofertilizer and biopesticide production sites and submit a report

Mangalore University

B. Sc – VI Semester, Botany Practical Examination

PLANT PHYSIOLOGY AND BIOTECHNOLOGY

Time: 3 hrs. Batch..... Date..... Time.....am/pm Max. Marks: 40

1. Conduct Major Experiment **A** 12 marks
2. Comment on minor experiments **B** and **C** 08 marks
3. Perform the Biochemical test **D** 04 marks
4. Identify and comment on **E** and **F** 06 marks
5. Practical Record and Field report 10 marks

1	A		Requirements - 02 Principle -01 Sketch-01 Procedure-02 Setting-04 Results and Inference-02 = 12	Register No. Assigned:
2	B		Procedure -02 Sketch -01	Absentees:
	C		Results and Inference <u>01</u> 04 each	Total Examined:
3	D		Procedure-02 Prep -01 Results and Inference-01	Examiners
4	E		Identification 01 Sk 01	1.
	F		Comment <u>01</u> 03 each	2.
5		Practical Record with field visit report	7+3=10	

Instruction to Examiners:

A Physiology Major Experiment by picking lots

B and **C** Physiology minor experiments

D- Biochemical test

E-Plant propagation methods/techniques; **F** -Any one biotechnological instrument

Program Name	BSc	Semester	IV/V/VI
Compulsory course	Skill Enhancement Program – Project work/ Field training		
Course Code	BSCBOPRS__	No. of credits	2
Contact hours	2hrs/week	Duration of SEA/Exam	3 hrs.
Formative assessment	10	Summative assessment	40

Proposed Topics for Project Work / Field Training (Botany)

1. Study of plant diversity in various habitats (gardens / croplands / grasslands / forests / wetlands / coastal areas).
2. Preparation of a checklist of medicinal plants of the local area.
3. Identification and documentation of ornamental plants.
4. Study of weed flora in agricultural fields.
5. Listing and identification of avenue trees.
6. Study of aquatic macrophytes in ponds / lakes / wetlands.
7. Identification and study of freshwater algae.
8. Identification and study of fungi and lichens from local habitats.
9. Survey and documentation of plant diseases in crops / horticultural plants.
10. Study of plant pests and associated symptoms.
11. Ethnobotanical survey of locally used plants.
12. Study of biodiversity in sacred groves.
13. Plant community analysis using quadrat method (frequency / density / abundance).
14. Study of leaf architecture and morphological variations.
15. Comparative study of floral morphology in selected taxa.
16. Study of seed germination under different environmental conditions.
17. Study of vegetative propagation techniques.
18. Study of soil characteristics influencing vegetation.
19. Documentation of endemic / rare / threatened plant species.
20. Study of pollination biology and floral visitors.
21. Study of urban flora / campus flora.
22. Study of economic botany (fibres / oils / spices / timber plants).
23. Evaluation of allelopathic effects of plant extracts.
24. Study of plant adaptations to abiotic stress (salinity / drought / flooding).
25. Assessment of biodiversity awareness and conservation attitudes among students / community.

Note

In addition to the above topics, projects addressing local botanical interest, ecosystem studies, environmental issues, conservation biology, sustainable utilization of plant resources, and climate change impacts or any topics relevant to Botany may be undertaken.

Ethical Guidelines

- Field studies should be conducted without disturbing natural habitats.
- Avoid unnecessary collection or destruction of plant specimens.
- Follow institutional and biodiversity conservation regulations.
- Obtain prior permission where required.

Guidelines for Project Work

1. Project allotment shall be done at the beginning of the semester.
2. Topic shall be finalized in consultation with the Project Guide.
3. Project work shall be carried out individually or in small groups (2–4 students).
4. Dissertation shall be submitted in the prescribed format.
5. Work shall include field/laboratory component, data analysis, and interpretation.
6. Submission must be certified by Project Guide and HOD.

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SCHEME OF EXAMINATION

B.Sc. BOTANY: IV/V/VI SEMESTER

COMPULSORY PAPER – SKILL ENHANCEMENT (PROJECT WORK / FIELD TRAINING)

Code: BSCBOPRS 602

Duration: 3 Hours

Max. Marks: 40

I. Field Work and Preparation of Dissertation 20 Marks

*(To be evaluated by Project Guide + Internal Examiner** + External Examiner**; each for 20 marks and the average shall be taken.)**

- The **Project Guide** shall assess the candidate based on his/her involvement in field/laboratory work and preparation of the dissertation. Marks allotment for the same shall be handed over to the **HOD in a sealed cover**, which will be transmitted to the examiners.

** Distribution of Marks:

- Introduction with Review of Literature – **3 Marks**
- Materials and Methods – **3 Marks**
- Result and Discussion – **8 Marks**
- Summary – **1 Mark**
- References – **1 Mark**
- Plates containing original photographs – **4 Marks**

II. Presentation by the Candidate* *(PPT slides preferably / charts)* 10 Marks

(Preparation of PPT slides / charts – 4 Marks; Presentation – 6 Marks)

*Presentation to be done **individually**, even if the project is carried out in a group.*

III. Viva-Voce *(Based on the contents of the dissertation)* 10 Marks

Note: Internal assessment marks shall be allotted based on the preparation and presentation of the dissertation topic.

Question Paper Pattern for Theory

Reg. No.

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BSCBO XX 000

__ Semester B. Sc. Degree Examination, _____ 202
(SEP) (2024-25 batch onwards)

BOTANY

Title of the Course

Time: 3 hrs.

Max. Marks: 80

Instructions: 1) Answer Part - A and Part – B

2) Answer any four full questions from Part- B choosing
one full question from each unit

3) **All** questions on Part – B carry **equal** marks

4) Draw diagrams **wherever** necessary

Part - A

I. Answer **any ten** of the following:

(10 x 2 = 20)

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

Part - B

Unit - 1

- | | | |
|-----|----|---|
| 13. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 14. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - II

- | | | |
|-----|----|---|
| 15. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 16. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - III

- | | | |
|-----|----|---|
| 17. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 18. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - IV

- | | | |
|-----|----|---|
| 19. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 20. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Question Paper Pattern for Theory

Reg. No.

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BSCBO XX 000

__ Semester B. Sc. Degree Examination, _____ 202
(SEP) (2024-25 batch onwards)

BOTANY

Title of the Course

Time: 3 hrs.

Max. Marks: 80

Instructions: 1) Answer Part - A and Part – B

2) Answer any four full questions from Part- B choosing
one full question from each unit

3) *All* questions on Part – B carry *equal* marks

4) Draw diagrams *wherever* necessary

Part - A

I. Answer **any ten** of the following:

(10 x 2 = 20)

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

Part - B

Unit - 1

- | | | |
|-----|----|---|
| 13. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 14. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - II

- | | | |
|-----|----|---|
| 15. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 16. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - III

- | | | |
|-----|----|---|
| 17. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 18. | a) | 4 |
| | b) | 4 |
| | c) | 7 |

Unit - IV

- | | | |
|-----|----|---|
| 19. | a) | 3 |
| | b) | 5 |
| | c) | 7 |

OR

- | | | |
|-----|----|---|
| 20. | a) | 4 |
| | b) | 4 |
| | c) | 7 |
